



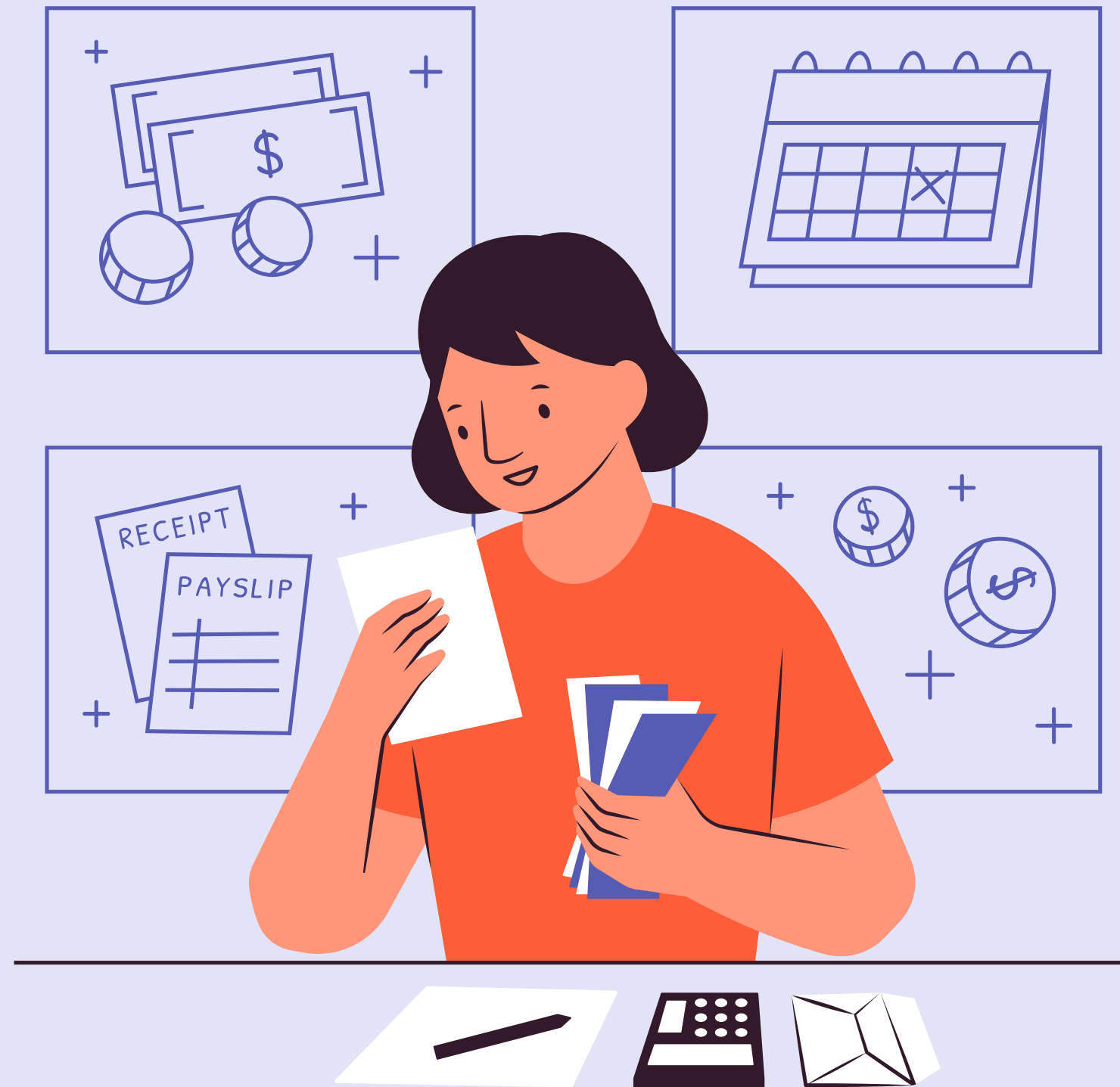
SALARY PREDICTOR WEB APP

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PROJECT OVERVIEW



- Flask web application to predict high-salary jobs
 - Based on company rating, founding year, seniority, state, and average salary
- Trained using scikit-learn models and feature selection

Step 1

CODE STRUCTURE



- Loaded dataset using Pandas and performed initial cleaning
- Extracted relevant columns: Rating, Founded, State, Seniority, avg_salary
 - Encoded categorical features using get_dummies
- Created binary target variable: high_salary = 1 if above median
 - Split data using train_test_split

FEATURE SELECTION

- Used SelectKBest for selecting top 5 features
- Score function: `f_classif` (ANOVA F-test)
- This is a filter-based method
 - Transformed training and testing data accordingly



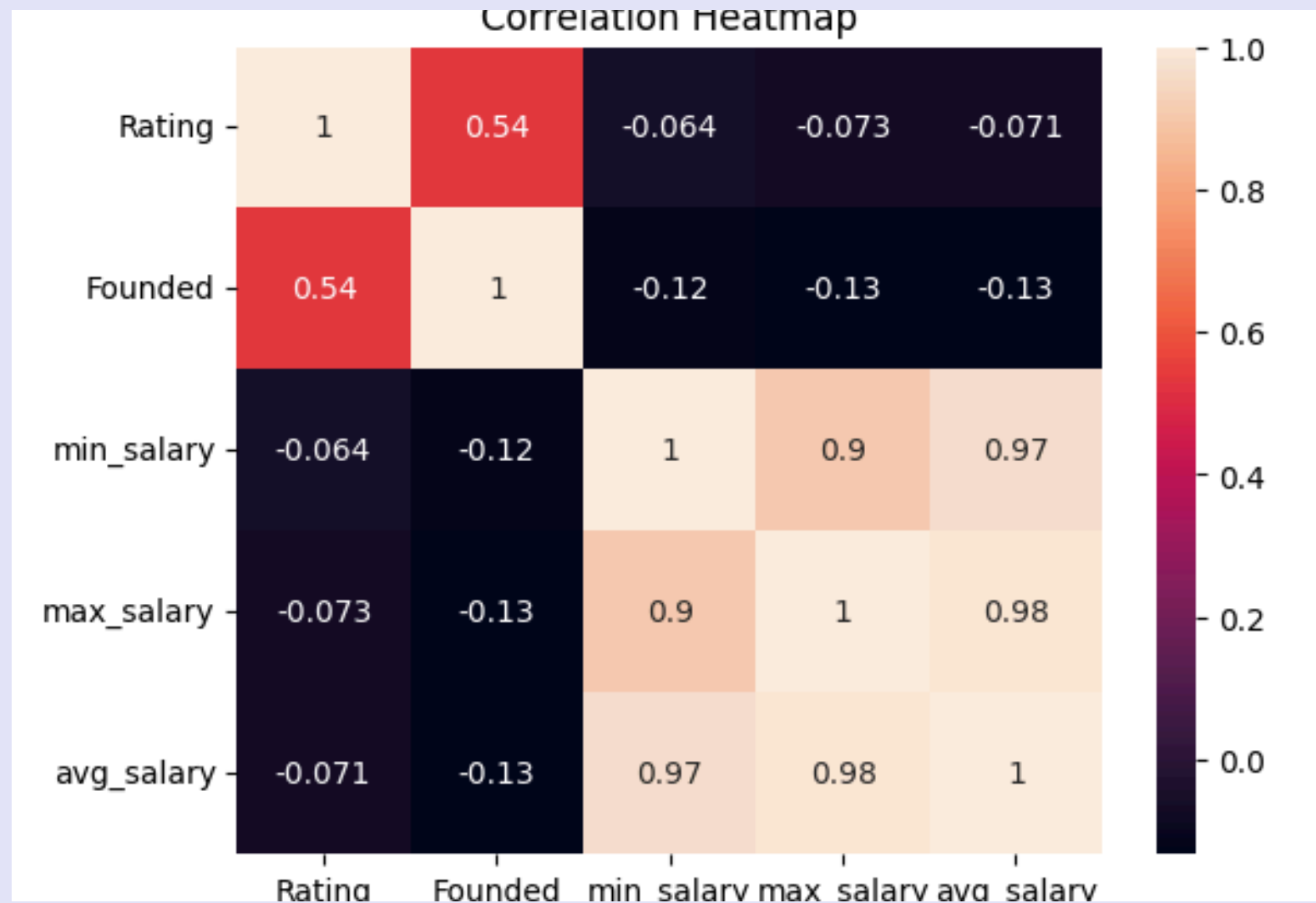


MODEL TRAINING AND EVALUATION

- Trained multiple classifiers: Logistic Regression, Random Forest, SVC
- Used `classification_report` to evaluate predictions
 - Selected best model based on performance
- Saved final model and feature list using `joblib`



CORRELATION HEATMAP

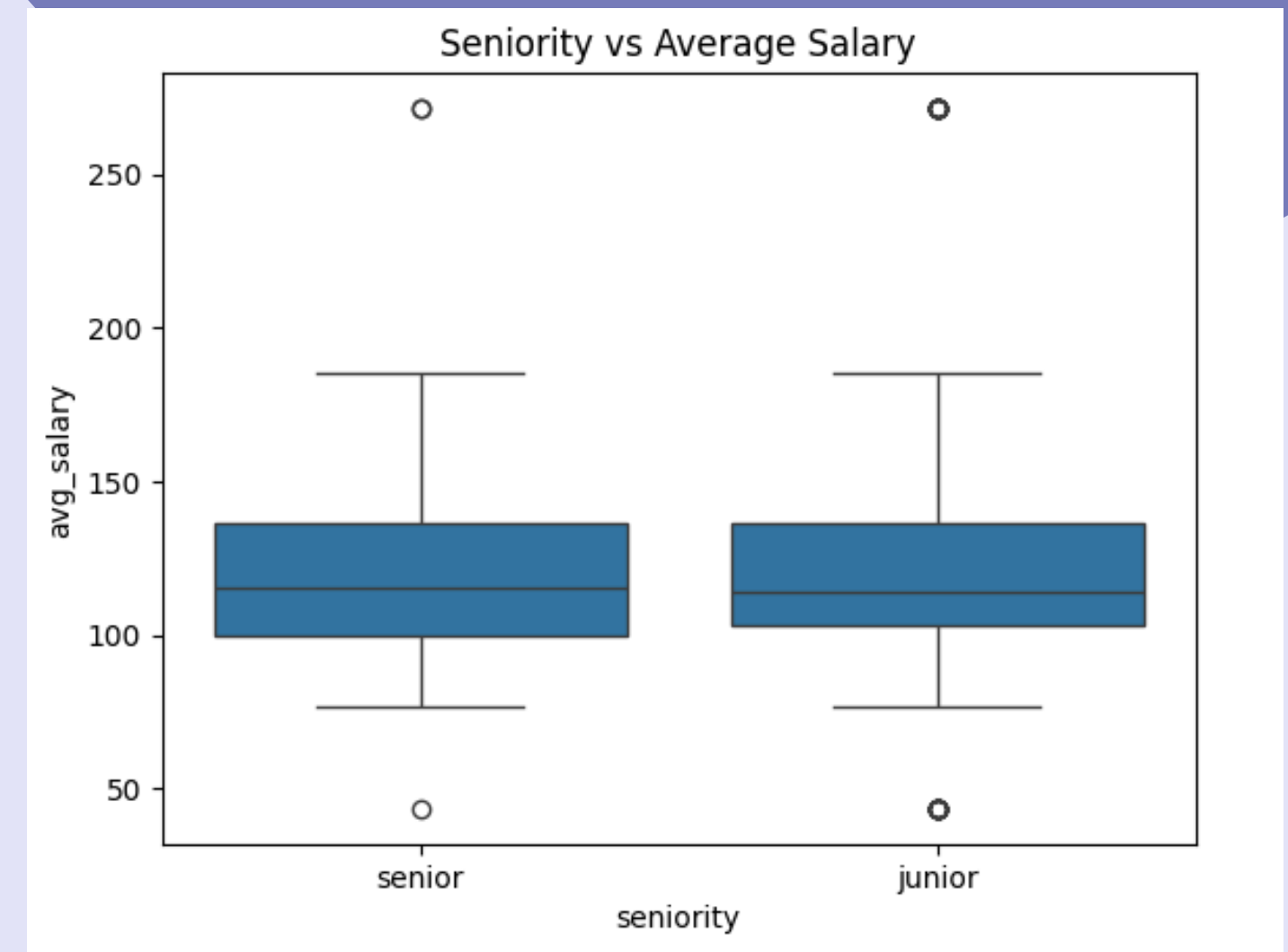


The heatmap illustrates the correlation between numerical features in the dataset. Strong correlations are found between min_salary, max_salary, and avg_salary. Rating and Founded year show weak correlation with salary, which suggests their limited predictive power.



SENIORITY VS. AVERAGE SALARY

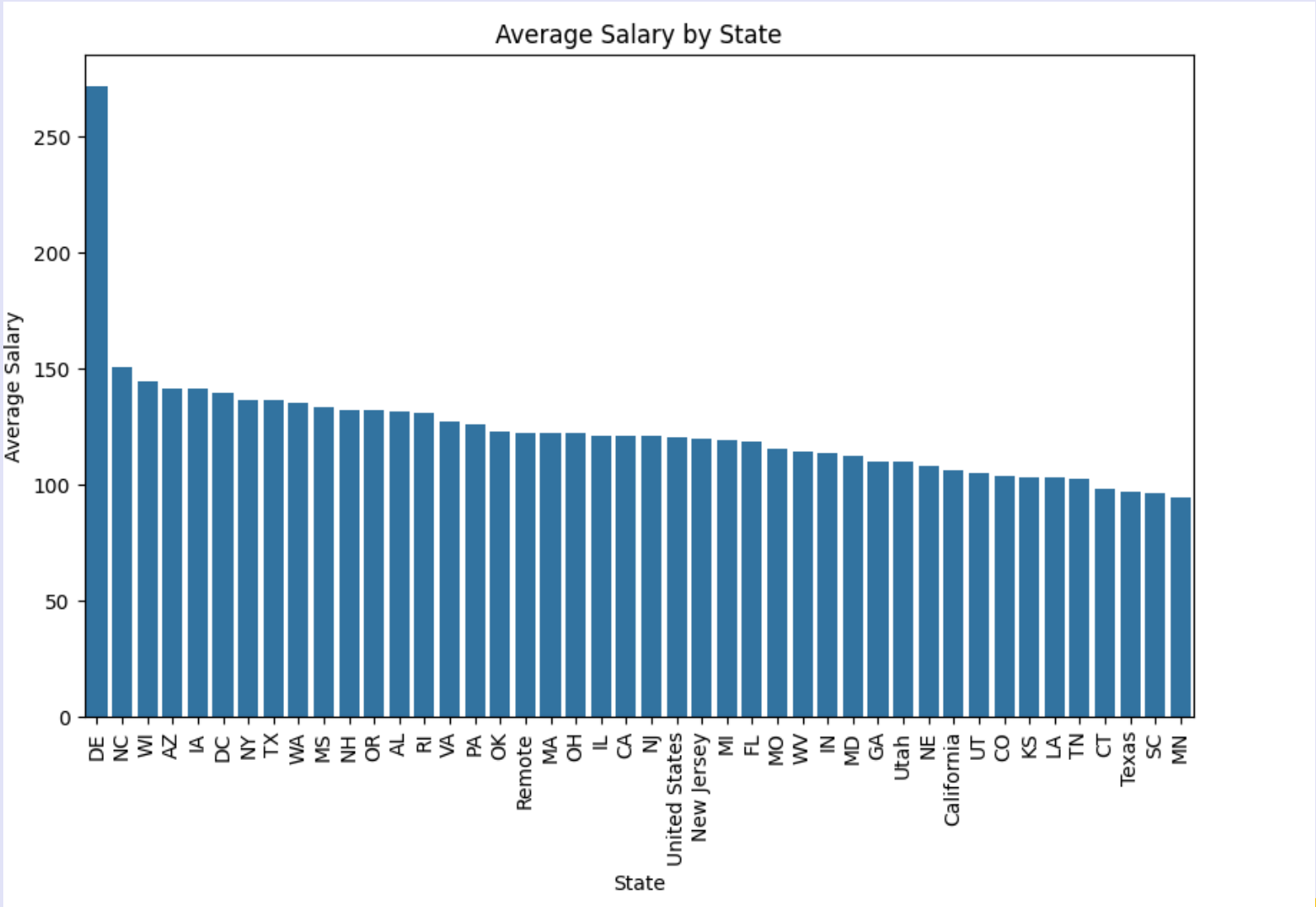
The boxplot compares the distribution of average salaries between junior and senior roles. Senior positions generally exhibit a higher average salary and wider salary spread. However, there is some overlap suggesting that other factors also influence salary.



AVERAGE SALARY BY STATE



THE BAR CHART ILLUSTRATES HOW AVERAGE SALARIES VARY ACROSS DIFFERENT STATES. DELAWARE AND NORTH CAROLINA LEAD WITH THE HIGHEST AVERAGE SALARIES. THIS ANALYSIS REVEALS GEOGRAPHICAL TRENDS IN SALARY EXPECTATIONS.



WEB APP



- Rating (float)
- Founded (year)
- Seniority (categorical)
 - State (categorical)
- Average Salary (float)

WEB APP



- index.html collects form input
- app.py receives and processes input
 - Predicts using model.pkl and features.pkl
- Returns result to HTML form

WEB APP



- Input:
 - Rating: 4.2
 - Founded: 2010
 - Seniority: Senior
 - State: CA
 - Avg Salary: 105000
- Output:
 - Prediction: High Salary

Predict if the Job is High Salary

Rating:
Founded:
Seniority:
State:
Average Salary:



THANK YOU!

